Phase 5: Comprehensive Requirements Catalog

Document Code: REQ-001

Project: Duplicate Bin Ecosystem

Methodology: IEEE 830 / ISO/IEC/IEEE 29148 Standard Requirements Engineering

Traceability Level: 100% Repository-Verified Evidence

1. Requirements Taxonomy & ID Nomenclature

- **BR-xxx:** Business Requirements (Executive & Organizational Goals)
- **SR-xxx:** Stakeholder Requirements (User Persona Expectations)
- **FR-xxx:** Functional Requirements (System Behaviors & Capabilities)
- **NFR-xxx:** Non-Functional Requirements (Performance, Usability, Reliability)
- **DR-xxx:** Data Requirements (Storage, Schemas & Persistence)
- **INT-xxx:** Integration Requirements (APIs, WebSockets & External Tools)
- **SEC-xxx:** Security & Compliance Requirements (Safety, Auth, Hardening)
- **REP-xxx:** Reporting & Telemetry Requirements (Dashboards, Logs, Export)

2. Business Requirements (BR)

Req ID	Requirement Statement	Business Objective	Evidence in Repository	Module	Priority	Dependencies	Status
BR-001	The system shall reduce consumed	OBJ-03 (Space Reclamation)	docs/SAAS_ANALYSIS_AND_STRATEGY.md#L	Core Engine	**Must Have**	None	**Implemented**

	consumed		EGY.md#L				

Req ID	Requirement Statement	Business Objective	Evidence in Repository	Module	Priority	Dependencies	Status
	storage volume on client systems by identifying and eliminating byte-identical and near-identical files.		9				
BR-002	The system shall provide an irreversible risk-free operational safety net ensuring zero permanent data loss during deduplicat	OBJ-02 (Zero Data Loss)	duplicate_bin/engine/archive.py#L20-L80	Archive System	**Must Have**	BR-001	**Implemented**

	deduplicat						

Req ID	Requirement Statement	Business Objective	Evidence in Repository	Module	Priority	Dependencies	Status
	ion operations.						
BR-003	The system shall support commercial hybrid SaaS monetization through Pay-As-You-Go deduplication credit packs and recurring team subscriptions.	OBJ-04 (Commercialization)	backend/app/main.py#L296-L318	SaaS Backend	**Must Have**	BR-001	**Implemented**
BR-004	The system shall enable corporate	OBJ-01 (Fleet Management)	backend/app/main.py#L245-L290	SaaS Management	**Should Have**	BR-003	**Implemented**

	corporate						

Req ID	Requirement Statement	Business Objective	Evidence in Repository	Module	Priority	Dependencies	Status
	IT administrators to monitor and orchestrate storage deduplication across distributed fleet workstations from a central portal.						

3. Functional Requirements (FR)

Req ID	Feature / Capability	Detailed Functional Requirement Statement	Evidence in Code	Module	Priority	Status
FR-001	Multi-Algorithm Hashing	The engine shall compute file content	duplicate_bin/core/hashing.py#L25-L65	Core Engine	**Must Have**	**Implemented**

		content				

Req ID	Feature / Capability	Detailed Functional Requirement Statement	Evidence in Code	Module	Priority	Status
		digests using SHA-256, MD5, CRC32, or BLAKE2b based on user configuration.				
FR-002	Two-Pass Size Pre-Filtering	The engine shall group candidate files by exact file size in bytes before computing cryptographic hashes, immediately discarding unique file sizes in $O(n)$ time.	duplicate_bin/core/detection.py#L85-L120	Detection Engine	**Must Have**	**Implemented**
FR-003	Union-Find Duplicate Grouping	The engine shall employ a Union-Find disjoint-set data	docs/TECHNICAL_ARCHITECTURE.md#L85-L89,	Detection Engine	**Must Have**	**Implemented**

		data	L89,			

Req ID	Feature / Capability	Detailed Functional Requirement Statement	Evidence in Code	Module	Priority	Status
		structure with path compression to accurately cluster transitive duplicate relationships ($A=B \land B=C \implies {A,B,C}$).	duplicate_bin/core/detection.py#L140			
FR-004	Automated Selection Heuristics	The system shall provide 8 distinct automated selection strategies (Keep Newest, Keep Oldest, Keep Largest, Keep Smallest, Keep Best Quality, Keep Deepest	duplicate_bin/core/selection.py#L15-L85	Selection Engine	**Must Have**	**Implemented**

		Deepest				

Req ID	Feature / Capability	Detailed Functional Requirement Statement	Evidence in Code	Module	Priority	Status
		Path, Keep Shortest Path, Custom Priority).				
FR-005	Pre-Scan Filter Rules	The system shall permit users to filter files by extension, minimum size, maximum size, modification date range, and regular expression patterns (capped at 512 chars).	duplicate_bin/core/filtering.py#L10-L75	Filter Engine	**Must Have**	**Implemented**
FR-006	Safe Archive Quarantine	When a user requests file deletion, the system shall move files into a	duplicate_bin/engine/archive.py#L30-L70	Archive Engine	**Must Have**	**Implemented**

		into a				

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Req ID	Feature / Capability	Detailed Functional Requirement Statement	Evidence in Code	Module	Priority	Status
		designated quarantine directory (`~/.duplicate_bin/archive`) preserving directory hierarchy.				
FR-007	Instant Undo / Restore	The system shall provide an undo mechanism (`Ctrl+Z`) restoring archived files back to their exact original disk locations with metadata preserved.	duplicate_bin/engine/archive.py#L85-L125	Archive Engine	**Must Have**	**Implemented**
FR-008	Drag-and-Drop Ingestion	The GUI shall support dragging and dropping	duplicate_bin/ui/main_window.py#L180-L210	PySide6 UI	**Should Have**	**Implemented**

		dropping				

Req ID	Feature / Capability	Detailed Functional Requirement Statement	Evidence in Code	Module	Priority	Status
		directory folders directly onto the application window to trigger an immediate scan configuration.				
FR-009	Side-by-Side Visual Previews	The system shall generate visual thumbnail previews and metadata side-by-side for candidate images, audio tags, and PDF documents.	duplicate_bin/ui/preview_widget.py#L25-L95	PySide6 UI	**Should Have**	**Implemented**
FR-010	Device 6-Digit Pairing	The system shall support	backend/app/	SaaS / Auth	**Must Have**	**Implemented**

		shall support	app/			

Req ID	Feature / Capability	Detailed Functional Requirement Statement	Evidence in Code	Module	Priority	Status
		pairing a desktop client daemon to a central cloud account using an expiring 6-digit cryptographic pairing code.	main.py#L74-L130			
FR-011	Remote Scan Dispatch	The SaaS backend shall allow users to trigger scans on active paired agents remotely via persistent WebSocket channels.	backend/app/main.py#L246-L290	SaaS / WS	**Should Have**	**Implemented**
FR-012	Credit Deduction Ledger	The SaaS backend shall deduct	backend/app/main.py#L16	Billing Engine	**Must Have**	**Implemented**

		shall deduct	main.py#L16			

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Req ID	Feature / Capability	Detailed Functional Requirement Statement	Evidence in Code	Module	Priority	Status
		deduplication credits (`credit_bytes`) from the user's account upon receiving verified scan telemetry indicating recovered bytes.	3-L198			
FR-013	13 Global Keyboard Shortcuts	The GUI shall provide full operational keyboard navigation for core actions (`Ctrl+O`, `Ctrl+S`, `Ctrl+F`, `Ctrl+Z`, `Ctrl+A`, `F1`, etc.).	duplicate_bin/ui/shortcuts.py#L10-L45	PySide6 UI	**Could Have**	**Implemented**
FR-014	Extensible	The core	duplicate_bi	Plugin	**Could Have**	**Implemente**

	Extensible	The core	duplicate_bi	Plugin		**Implemente**

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Req ID	Feature / Capability	Detailed Functional Requirement Statement	Evidence in Code	Module	Priority	Status
	Plugin System	engine shall expose an event hook lifecycle (`on_scan_start`, `on_scan_complete`, `on_file_archived`) allowing external modular extensions.	n/plugins/manager.py	System		**d**

4. Non-Functional Requirements (NFR)

Req ID	Category	Requirement Statement	Target Metric / Benchmark	Evidence	Priority
NFR-001	Performance (Scan Velocity)	The system shall complete initial scanning of 10,000 files in under 20 seconds on standard multi-	18.7 seconds verified	README.md#L231	**Must Have**

		standard multi-			

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Req ID	Category	Requirement Statement	Target Metric / Benchmark	Evidence	Priority
		core hardware.			
NFR-002	Performance (Cache Rescan)	The system shall complete rescanning of an unchanged 10,000-file directory in under 5 seconds using persistent hash caching.	4.2 seconds verified (3-5x speedup)	README.md#L232	**Must Have**
NFR-003	Scalability & Memory Bounding	The engine shall use streaming chunk processing (4KB to 1MB) ensuring client RAM consumption remains bounded (< 200 MB for 10k files, < 850 MB for 100k files).	Max 850 MB on 100k files	README.md#L233, performance_engine.py#L65	**Must Have**
NFR-004	Usability (UI Responsiveness)	Long-running file scans and hash computations shall execute on decoupled	Worker decoupling via Qt Signals	duplicate_bin/ui/scan_worker.py#L15-L60	**Must Have**

		decoupled			

Req ID	Category	Requirement Statement	Target Metric / Benchmark	Evidence	Priority
		worker threads (`QThread`), maintaining 60 FPS GUI responsiveness with zero freezing.			
NFR-005	Reliability & Crash Recovery	The system shall record JSON scan checkpoints periodically, allowing interrupted scans to resume without re-processing already hashed items.	JSON Checkpoint Manager	duplicate_bin/engine/checkpoint.py#L20-L75	**Should Have**
NFR-006	Portability & Cross-Platform	The application shall package natively for Linux (.deb), Windows (.exe standalone), and macOS without environment configuration.	PyInstaller spec & Debian scripts	build_windows.py, build_debian.sh	**Must Have**

5. Security Requirements (SEC)

Req ID	Requirement Statement	Business Rationale	Evidence	Priority
SEC-001	System Directory Protection	The system shall refuse to scan or modify protected operating system roots (`C:\Windows`, `C:\Program Files`, `/bin`, `/usr`, `/etc`, `/System`).	Prevent catastrophic operating system corruption.	duplicate_bin/core/utils.py#L15-L45
SEC-002	Path Traversal & Symlink Safety	The archive system shall resolve absolute canonical paths (`os.path.realpath`) to verify that no target destination escapes the designated sandbox boundary.	Prevent directory traversal exploits (`../../etc`).	docs/TECHNICAL_ARCHITECTURE.md#L141-L142
SEC-003	Safe Checkpoint Deserialization	Checkpoint data shall be persisted strictly using JSON formats; arbitrary object deserialization (`pickle`) is prohibited.	Prevent remote code execution via poisoned checkpoint files.	docs/TECHNICAL_ARCHITECTURE.md#L100-L103
SEC-004	API Authentication & Token Security	Cloud API endpoints shall mandate OAuth2 Bearer JWT	Prevent unauthorized access to device commands	backend/app/auth.py#L15-L55

		OAuth2 Bearer JWT	to device commands	

Req ID	Requirement Statement	Business Rationale	Evidence	Priority
		authentication (HS256) with password storage hashed via Bcrypt.	and telemetry.	

6. Data & Integration Requirements (DR / INT)

Req ID	Requirement Statement	Source / Schema	Target Entity	Priority
DR-001	Persistent Hash Cache Database	SQLite schema (`hash_cache.db`) storing `(file_path, file_size, mtime, hash_value, algorithm, created_at)` with WAL mode.	`PersistentHashCache`	**Must Have**
DR-002	SaaS User & Device Ledger	Relational schema modeling `users`, `devices`, and `scans` with foreign keys and cascade delete rules.	`backend/app/models.py`	**Must Have**
INT-001	WebSocket Agent Relay Channel	Bidirectional real-time channel (`/ws/agent`) transmitting status payloads (`AGENT_PROGRESS`, `AGENT_COMPLETE`, `CMD_START_SCAN`).	`websocket_manager.py`	**Must Have**

Req ID	Requirement Statement	Source / Schema	Target Entity	Priority
INT-002	REST Telemetry Sync	Endpoint (`POST /api/scans/telemetry`) ingesting scan metrics (`bytes_recoverable`, `duplicates_count`, `duration_seconds`).	`backend/app/main.py#L149`	**Must Have**

7. Reporting Requirements (REP)

Req ID	Requirement Statement	Output Format	Evidence	Priority
REP-001	Scan Audit Report Generation	The system shall export comprehensive duplicate analysis reports detailing group clusters, file paths, hashes, and size savings.	HTML, JSON, CSV formats	duplicate_bin/engine/reporting.py#L25-L120
REP-002	Central Web Dashboard Analytics	The SaaS portal shall aggregate and display total bytes saved, active paired devices, remaining credit balance, and scan history.	JSON response to UI (`/api/dashboard/stats`)	backend/app/main.py#L202-L243